PRIVATE CYBER LAB HANDBOOK

LAB 04 — USER AND GROUP MANAGEMENT IN ACTIVE DIRECTORY

Windows Server 2019 | Active Directory Administration | Private Cybersecurity Lab

Author: [Your Name]

Domain Controller: DC01

Operating System: Windows Server 2019

Domain: zroots.local (DC=zroots,DC=local)

Tool: PowerShell (ActiveDirectory Module)

Status: Completed & Verified

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1.0 Lab Overview and Objectives

This document records the execution of **Lab 04: User and Group Management in Active Directory** within a private VMware cybersecurity lab environment running Windows Server 2019. Proper identity and access management (IAM) forms the bedrock of enterprise Active Directory security. This lab establishes core administrative competency in creating, configuring, auditing, and maintaining directory objects using native Windows PowerShell Active Directory modules.

> **Primary Objectives:**
>
> - Inventory and inspect existing user accounts and security groups within `zroots.local`.
> - Perform granular group membership assignments following the principle of least privilege.
> - Modify user attributes (such as department and office locations) programmatically.
> - Execute account lifecycle operations (disable, enable, and password policy checks).
> - Troubleshoot domain password complexity constraints and command execution hygiene.

2.0 Lab Topology and Prerequisites

The lab is hosted on an isolated private VMware virtual machine network. The primary domain controller, **DC01**, hosts the `zroots.local` forest and domain root. All administrative operations are conducted via Windows PowerShell leveraging the built-in `ActiveDirectory` module.

3.0 User Account Inventory

Initial discovery identified standard and administrative user accounts within the directory structure:

Account Name	Distinguished Name (DN)	Enabled	Primary Role
Administrator	CN=Administrator,CN=Users,DC=zroots,DC=local	True	Built-in Domain Admin
Guest	CN=Guest,CN=Users,DC=zroots,DC=local	False	Built-in Guest Account
krbtgt	CN=krbtgt,CN=Users,DC=zroots,DC=local	False	Key Distribution Center Service
[Test User]	CN=[User],OU=Employees,DC=zroots,DC=local	True	Lab Testing Subject

4.0 Security Group Inventory

Active Directory security groups govern authorization scopes across the domain. The environment utilizes Global Security groups for role-based permissions:

Group Name	Scope	Category	Description / Purpose
Domain Admins	Global	Security	Full administrative control over domain
Domain Users	Global	Security	Default security group for all domain accounts
IT-Operators	Global	Security	Delegated operational privileges

5.0 Group Membership Management

To assign permissions securely, user accounts are added to specific security groups using the `Add-ADGroupMember` cmdlet. The command below adds the target lab user to the `IT-Operators` security group:

```
Add-ADGroupMember -Identity "IT-Operators" -Members "TestUser"
```

Command Breakdown:

- `Add-ADGroupMember`: Adds one or more objects to an Active Directory group.
- `-Identity`: Specifies the target group by name, SID, or Distinguished Name.
- `-Members`: Specifies the user account object(s) to be added.

6.0 User Attribute Modification

Active Directory user attributes store metadata critical for organizational governance. The following cmdlet updates the employee's department and office location attributes:

```
Set-ADUser -Identity "TestUser" -Department "Cybersecurity" -Office "Server Room 01"
```

Verification via `Get-ADUser -Identity "TestUser" -Properties Department, Office` confirmed successful attribute population across the directory partition.

7.0 Account Disable and Re-Enable Exercise

Account lifecycle management ensures inactive or compromised accounts can be immediately locked out without deleting their security history or SID binding. The lab verified account disabling and re-enabling using the `Set-ADUser` cmdlet with the `-Enabled` boolean parameter.

```
      Set-ADUser -Identity "TestUser" -Enabled $false Set-ADUser -Identity "TestUser" -
Enabled $true
```

8.0 Domain Password Policy Inspection

Domain security settings dictate password complexity and rotation schedules. Inspection via `Get-ADDefaultDomainPasswordPolicy` revealed the following active domain configuration:

Policy Property	Observed Value	Security Assessment
Minimum Password Length	7 characters	Below modern baseline (NIST recommends 8+)
Complexity Enabled	True	Enforces uppercase, lowercase, numbers, symbols
Password History	24 passwords remembered	Prevents immediate password reuse
Maximum Password Age	42 days	Standard rotation window

9.0 Test User Password Reset

Administrative password resets require secure handling via the Account Management API. In PowerShell, this is accomplished using a secure string object passed to `Set-ADAccountPassword`.

> **Observed Behavior during Password Reset:**
> The first password reset attempt failed because the supplied test string did not meet the domain complexity or length criteria (minimum 7 characters). Active Directory rejected the operation with a policy violation error. Upon supplying a compliant password, the second attempt succeeded immediately.

10.0 Troubleshooting and Error Analysis

Two primary administrative hurdles were documented during execution:

- **Password Policy Rejection:** Initial testing with a short password string threw an AD constraint violation. *Resolution:* Evaluated policy via `Get-ADDefaultDomainPasswordPolicy` and formulated a compliant string.
- **PowerShell Paste Artifacts:** Accidental pasting of prompt symbols (`PS C:\>`) caused syntax exceptions. *Resolution:* Flushed the buffer and executed clean command strings.

11.0 Security Lessons and Best Practices

Identity administration requires rigorous adherence to security principles:

- **Least Privilege:** Never add standard user accounts to privileged groups like `Domain Admins` unless strictly necessary for tier-zero administration. Use scoped security groups like `IT-Operators`.
- **Attribute Hygiene:** Keeping user attributes (department, manager, office) accurate aids in access review audits and rapid identification of orphaned accounts.
- **Authentication vs. Authorization:** A command success (e.g., password reset) proves that the administrative action took place in the directory store, but it does not test end-to-end Kerberos ticket granting or interactive workstation logon unless explicitly performed.

12.0 Knowledge-Check Questions & Reference Answers

Q1: Why is disabling an account preferable to deleting it when an employee leaves?
Answer: Deleting an account destroys its SID history and object GUID, which can orphan file permissions and auditing logs. Disabling preserves security associations while instantly revoking access.

Q2: What Active Directory component enforces minimum password length across all domain users?
Answer: The Default Domain Password Policy (managed via Group Policy and AD objects).

13.0 Evidence and Screenshot Checklist

Verification Item	Target Object	Status
Domain Controller Connectivity	DC01 (zroots.local)	Passed
User Account Inventory Query	Get-ADUser	Passed
Group Membership Assignment	IT-Operators	Passed
User Attribute Modification	Department / Office	Passed
Password Policy Inspection	Default Domain Policy	Passed

14.0 Appendix: Complete PowerShell Command Reference

```
      # 1. Inspect Domain Controller and AD Module Import-Module ActiveDirectory Get-AD
Domain # 2. Query User and Group Inventories Get-ADUser -Filter * -Properties Departmen
t, Office Get-ADGroup -Filter * # 3. Modify Group Membership and Attributes Add-ADGroup
Member -Identity "IT-Operators" -Members "TestUser" Set-ADUser -Identity "TestUser" -De
partment "Cybersecurity" -Office "Server Room 01" # 4. Account Lifecycle Management Set
-ADUser -Identity "TestUser" -Enabled $false Set-ADUser -Identity "TestUser" -Enabled
$true # 5. Inspect Password Policy Get-ADDefaultDomainPasswordPolicy
```